



Unit 3 · Outcome 1 · Maintaining Biodiversity

Why genetic diversity matters when the environment changes

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 02 Why genetic diversity matters when the environment changes A population can only survive a new threat using alleles it already carries

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Why genetic diversity matters when the environment changes

UNIT 3 · AOS 1 · KEY KNOWLEDGE 02 Why genetic diversity matters when the environment changes A population can only survive a new threat using alleles it already carries . Genetic diversity is that hidden reserve — and once it's gone, it's almost impossible to rebuild. scenario: Kestrel Ridge Reserve · box-ironbark woodland, central Victoria

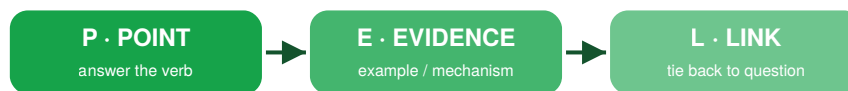
Key comparison tables

Term	What it means	Kestrel Ridge example
Gene	A length of DNA coding for a characteristic.	The gene influencing immune response in honeyeaters.
Allele	One of the alternative versions of a gene.	An allele giving resistance to a beak-and-feather virus.
Gene pool	All the alleles present in all individuals of a population.	Every allele carried by the ~250 captive + wild Helmeted Honeyeaters.
Allele frequency	How common an allele is, as a proportion of the gene pool.	If 30 of 200 copies are the resistance allele → frequency 0.15.
Heterozygosity	Proportion of individuals carrying two different alleles at a gene — a common measure of genetic diversity.	High heterozygosity = a healthier, more varied population.

Force	What happens	Effect on diversity
Bottleneck	A sharp crash (fire, clearing, disease) kills most individuals; survivors are a random, small sample.	Many alleles lost permanently.
Founder effect	A few individuals start a new isolated population, carrying only a fraction of the original alleles.	New group is genetically narrow from day one.
Genetic drift	In small populations, allele frequencies change by chance alone each generation; rare alleles easily vanish.	Diversity erodes randomly.
Inbreeding	Related individuals mate (unavoidable when few remain), raising homozygosity and exposing harmful recessive alleles.	Inbreeding depression : lower fertility, survival & health.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Genetic diversity is when there are lots of different animals and plants in an area. Variation comes from animals breeding. Why it loses marks: describes species diversity, not genetic diversity within a population. "Breeding" recombines existing alleles — it isn't a source of new variation.

✓ STRONG / MODEL ANSWER

Genetic diversity is the variety of alleles in the gene pool of a population or species. A source of new genetic variation is mutation, which creates entirely new alleles. Why it scores: precise allele-level definition + correctly identifies mutation as the source of new alleles (not recombination).

✗ WEAK ANSWER

A population with high genetic diversity is stronger so it can fight off disease better. The animals will adapt to the disease and become immune, then survive and not go extinct. Why it loses marks: "stronger" is vague; "animals adapt to become immune" is the Lamarckian error — individuals don't adapt on demand. No mention of alleles, selection or allele frequency.

✓ STRONG / MODEL ANSWER

High genetic diversity means many different alleles exist in the gene pool. A new disease acts as a selection pressure. Because more alleles are present, it is more likely that some individuals already carry an allele giving resistance. Those individuals survive and reproduce, passing the resistance allele on, so its frequency rises and the population recovers. A low-diversity population may contain no resistant individuals, so it cannot respond and may collapse. Why it scores: alleles → selection pressure → pre-existing resistance → survival & reproduction → frequency shift → contrast with low-diversity outcome. Full causal chain.

✗ WEAK ANSWER

When there are not many birds left there is less diversity because there are fewer of them. This is bad for the birds. Why it loses marks: circular ("fewer = less diversity") with no mechanism (drift/inbreeding). "Bad for the birds" is not a named consequence.

✓ STRONG / MODEL ANSWER

In a small population, mating between related individuals (inbreeding) is unavoidable and genetic drift causes allele frequencies to change randomly, so rare alleles are lost and heterozygosity falls — reducing genetic diversity. A consequence is inbreeding depression: more harmful recessive alleles are expressed, lowering fertility and survival. Why it scores: names two mechanisms (inbreeding, drift) with their effect on alleles, then a specific consequence (inbreeding depression with examples).

✗ WEAK ANSWER

They added new birds so there are more birds now. More birds is better because the population is bigger and won't go extinct as easily. Why it loses marks: treats it as just "more birds" (population size), missing the genetic point — new alleles. No link to future environmental change or adaptive potential, so the "justify" is unaddressed.

✓ STRONG / MODEL ANSWER

Introducing individuals from another lineage causes gene flow: new alleles enter the gene pool, raising genetic diversity and heterozygosity and reducing inbreeding. This is justified because a more varied gene pool has greater adaptive potential — if the environment changes in future (e.g. a new disease or warming climate), it is more likely that some individuals already carry a suitable allele, so the population can be selected to adapt rather than collapse. Why it scores: mechanism (gene flow → new alleles → diversity up, inbreeding down) + justification tied directly to future environmental change and adaptive potential.

✗ WEAK ANSWER

Population A will probably survive because it has more diversity and Population B will die because it has less. Diversity is good for populations and helps them survive diseases and other problems in the environment. Why it loses marks: states the outcome but never explains why via alleles/selection. No real comparison structure, no adaptive-potential mechanism, no mention of the bottleneck's effect. Generic.

✓ STRONG / MODEL ANSWER

Population A's high genetic diversity means its gene pool contains many different alleles, so it is likely that some frogs already carry an allele conferring resistance to the fungus. The pathogen acts as a selection pressure: resistant individuals survive and reproduce, the resistance allele's frequency rises over generations, and the population adapts and persists — it has high adaptive potential. Population B passed through a bottleneck, so genetic drift and the small surviving sample removed most of its allelic variation; heterozygosity is low and inbreeding is likely. It is therefore much less likely that any individual carries a resistance allele. With little variation for selection to